$D_{x,y} = \{(x^{(i)}, y^{(i)})\}_{i=1}^{N}$ $\quad x^{(i)} \in \mathbb{R}^d$, $y^{(i)} \in \{-1, 1\}$

Logistic Regression as a binary classifier.

$$P(y^{(i)} = 1 \mid x^{(i)}) = \sigma(w^T x + b)$$

$w \in \mathbb{R}^d$
$b \in \mathbb{R}$

Geometry of a linear classifier.

$x_2 = m x_1 + c \Rightarrow m x_1 - x_2 + c = 0$

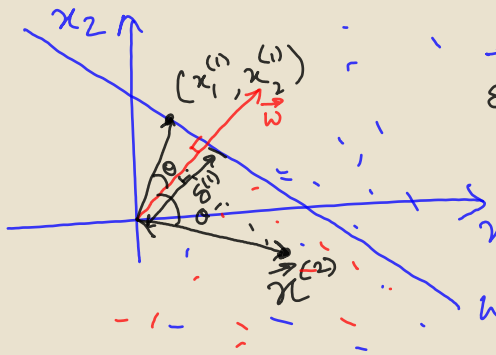
$[m \; -1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + c = 0$

$w_1 = m$
$w_2 = -1$
$b = c$

} eqn of line $\vec{w}^T \vec{x} + b = 0$

$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} m \\ -1 \end{bmatrix}$

$b = c$



$w^T x^{(1)} + b = 0$

$\delta^{(1)} = \|w\| \|x^{(1)}\| \cos\theta$

Assume $\|w\| = 1$

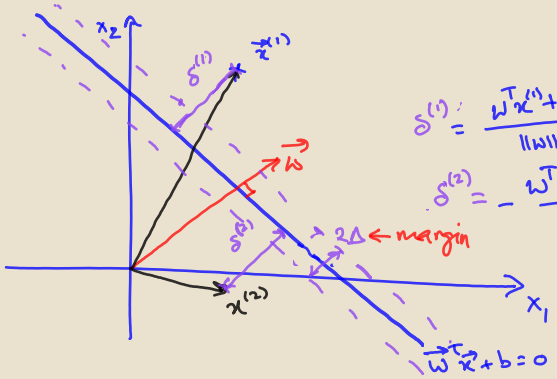
$\delta^{(2)} = \|w\| \|x^{(2)}\| \cos\theta'$

} geometric distance of points from the line

$w_1 x_1 + w_2 x_2 + b = 0 = \vec{w}^T \vec{x} + b$

$$\sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}}$$

Maximum margin Classifier



$\delta^{(1)} = \frac{w^T x^{(1)} + b}{\|w\|}$

$\delta^{(2)} = -\frac{w^T x^{(2)} + b}{\|w\|}$

$$\delta^{(i)} = y^{(i)} \frac{w^T x^{(i)} + b}{\|w\|}$$

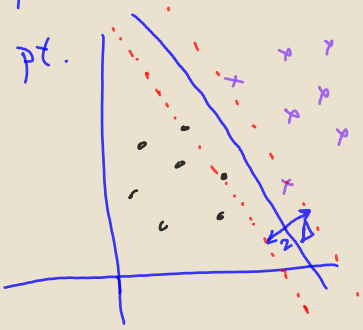
$(w^T x + b - \Delta) / \|w\|$
$(w^T x + b + \Delta) / \|w\|$

Margin maximization problem for binary classification

$$\max_{w,b} \; \Delta$$
$$\text{s.t. } y^{(i)}(w^T x^{(i)} + b) \geq \Delta \quad i = 1, \ldots, N$$
$$\|w\|_2 = 1$$

What is Δ?

The margin is the distance to the nearest point to the boundary.



As the boundary changes, so does the nearest pt. and therefore the margin.

$\therefore$ the margin is a function of the decision boundary parametrized by $(\vec{w}, b)$

i.e $$\Delta(\vec{w}, b) = \min_{i=1,\cdots,N} \delta^{(i)}(\vec{w}, b)$$

where $$\delta^{(i)} = \frac{\vec{w}^T \vec{x}^{(i)} + b}{\|\vec{w}\|}$$

Note. We can observe that $(\vec{w}, b)$ can be arbitarily scaled and it still represents the same hyperplane & therefore the same physical (or geometric) distance $\delta^{(i)}$

In other words

$$\delta^{(i)}(\vec{w}, b) = \frac{\vec{w}^T x^{(i)} + b}{\|\vec{w}\|} = \frac{(2\vec{w})^T \vec{x} + 2b}{\|2\vec{w}\|} = \frac{(c\vec{w})^T x^{(i)} + cb}{\|c\,\vec{w}\|} \quad \forall c > 0$$

Since the scaling factor can be arbitrarily chosen, for a given hyperplane, without changing the geometric distance of the points from the hyperplane, we can fix a scale arbitrarily.

By choosing the following strategy for scaling (w, b), we will see that our complicated optimization problem for margin maximization elegantly transforms into a convex optimization problem.

Consider an arbitrary hyperplane $(c\cdot\vec{w}', c\cdot b')_{c>0}$, we choose a scale $c = c^*$ such that

(I) $$\left\{ \min_{i=1,\cdots,N} \delta^{(i)}(c^*\vec{w}', c^* b') = \Delta(c^*\vec{w}', c^* b') = \frac{1}{\|c^*\vec{w}'\|} = \frac{1}{\|\vec{w}\|} \right.$$

where we write $\vec{w} = c^*\vec{w}'$ & $b = c^* b'$ as the fixed scale representation of the hyperplane.

We will come back to this.

Note that fixing the scale does not change the physical distance or the geometry AT ALL! It changes the algebraic expression & therefore the optimization problem.

Recall our naive max-margin optimization problem was

$$\max_{\vec{w},b} \quad \Delta(\vec{w},b)$$
$$\text{s.t.} \quad y^{(i)}(\vec{w}^T x^{(i)} + b) \geq \Delta(\vec{w},b) \qquad i = 1, \ldots, N$$
$$\|w\| = 1$$

← This is non-convex & we wish to remove it.

Rewriting,

$$\max_{\vec{w},b} \quad \Delta(\vec{w},b)$$
$$\text{s.t.} \quad \frac{y^{(i)}(\vec{w}^T x^{(i)} + b)}{\|\vec{w}\|} \geq \Delta(\vec{w},b) \qquad i = 1, \ldots, N$$

→ still the geometric distance of $x^{(i)}$ from the decision boundary

This problem should give us the same solution as the above problem but without the "$\|\vec{w}\|_2 = 1$" constraint. However the constraint is still non-convex. Moving on, we write

$$\max_{\vec{w},b} \quad \Delta(\vec{w},b)$$
$$\text{s.t.} \quad y^{(i)}(\vec{w}^T x^{(i)} + b) \geq \Delta(\vec{w},b) \cdot \|\vec{w}\| \qquad i = 1, \ldots, N$$

Now, recall our scale factor argument from eqn. (1) above.

If we fix a scale, such that

$$\Delta(\vec{w},b) = \frac{1}{\|\vec{w}\|}$$

Substituting, we get

$$\max_{\vec{w},b} \quad \frac{1}{\|\vec{w}\|}$$
$$\text{s.t.} \quad y^{(i)}(\vec{w}^T x^{(i)} + b) \geq \frac{1}{\|\vec{w}\|} \cdot \|\vec{w}\| \qquad i = 1, \ldots, N$$

$$\Rightarrow \quad \max_{\vec{w},b} \quad \frac{1}{\|\vec{w}\|}$$
$$\text{s.t.} \quad y^{(i)}(\vec{w}^T x^{(i)} + b) \geq 1 \qquad i = 1, \ldots, N$$

The constraints are now convex, but the objective

function is still non-convex However, maximizing $\frac{1}{\|\vec{w}\|}$ is the same as minimizing $\|w\|$ or for that matter $\|w\|_2^2$. The above problem can finally be written as

$$\min_{\vec{w},b} \|w\|_2^2$$
$$\text{s.t. } y^{(i)}(\vec{w}^T x^{(i)} + b) \geq 1 \qquad i = 1, \ldots, N$$

OR

$$\min_{\vec{w},b} \frac{1}{2} w^T w$$
$$\text{s.t. } y^{(i)}(\vec{w}^T x^{(i)} + b) \geq 1 \qquad i = 1, \ldots, N$$

This is a convex optimization problem that will give an optimal solution for a maximum margin binary classifier.